

MOBAN — Exam June 2025

Exercise 1: Linear Programming — Simplex Algorithm

Problem

$$\begin{aligned}
 \max z &= x_1 + 3x_2 - 4x_3 + x_4 \\
 \text{s.t.} \quad x_1 + x_2 &\leq 7 \\
 &\quad \quad x_3 \leq 9 \\
 x_1 - x_2 + x_4 &\leq 5 \\
 x_1, x_2, x_3, x_4 &\geq 0
 \end{aligned}$$

Step 1: Convert to Standard Form

Add slack variables $s_1, s_2, s_3 \geq 0$ to turn each inequality into an equality:

$$\begin{aligned}
 x_1 + x_2 &\quad + s_1 &= 7 \\
 &\quad x_3 + s_2 &= 9 \\
 x_1 - x_2 + x_4 &\quad + s_3 &= 5
 \end{aligned}$$

Initial Basic Feasible Solution (BFS):

- Basic variables: $s_1 = 7, s_2 = 9, s_3 = 5$
- Non-basic ($= 0$): $x_1 = 0, x_2 = 0, x_3 = 0, x_4 = 0$
- $z = 0$

Write the objective row as: **$z - x_1 - 3x_2 + 4x_3 - x_4 = 0$**

(Positive z-row coefficient = increasing that variable hurts $z \rightarrow$ don't enter. Negative = beneficial $\rightarrow$ enter.)

Note on x_3 : Its z-row coefficient is already **+4** in the initial tableau $\rightarrow$ increasing x_3 decreases $z \rightarrow x_3 = 0$ throughout all iterations.

Initial Tableau

BV	x_1	x_2	x_3	x_4	s_1	s_2	s_3	RHS
s_1	1	1	0	0	1	0	0	7
s_2	0	0	1	0	0	1	0	9
s_3	1	-1	0	1	0	0	1	5

BV	x1	x2	x3	x4	s1	s2	s3	RHS
z	-1	-3	4	-1	0	0	0	0

Iteration 1

Entering variable: x2 — most negative z-row coefficient (-3)

Min-ratio test (only rows with positive coefficient in the x2 column):

Row	RHS	x2 coeff	Ratio
s1	7	1	$7/1 = 7 \leftarrow \text{min}$
s2	9	0	— (skip)
s3	5	-1	— (negative, skip)

Leaving variable: s1 | **Pivot element:** row 1, column x2, value = 1

Row operations:

```

New x2-row   = Row1 ÷ 1           (no change, pivot = 1)
New s2-row   = Row2 - 0 × Row1    (no change, x2 coeff = 0)
New s3-row   = Row3 - (-1) × Row1 = Row3 + Row1
New z-row    = z-row - (-3) × Row1 = z-row + 3 × Row1
  
```

Tableau after Iteration 1:

BV	x1	x2	x3	x4	s1	s2	s3	RHS
x2	1	1	0	0	1	0	0	7
s2	0	0	1	0	0	1	0	9
s3	2	0	0	1	1	0	1	12
z	2	0	4	-1	3	0	0	21

Current BFS: x2 = 7, s2 = 9, s3 = 12, **z = 21**

Iteration 2

Entering variable: x4 — only remaining negative z-row coefficient (-1)

Min-ratio test:

Row	RHS	x4 coeff	Ratio
x2	7	0	— (skip)

Row	RHS	x4 coeff	Ratio
s2	9	0	— (skip)
s3	12	1	$12/1 = 12 \leftarrow$ only option

Leaving variable: s3 | **Pivot element:** row 3, column x4, value = 1

Row operations:

```

New x4-row   = Row3 ÷ 1           (no change, pivot = 1)
New x2-row   = Row1 - 0 × Row3    (no change, x4 coeff = 0)
New s2-row   = Row2 - 0 × Row3    (no change, x4 coeff = 0)
New z-row    = z-row - (-1) × Row3 = z-row + Row3

```

Tableau after Iteration 2 — OPTIMAL:

BV	x1	x2	x3	x4	s1	s2	s3	RHS
x2	1	1	0	0	1	0	0	7
s2	0	0	1	0	0	1	0	9
x4	2	0	0	1	1	0	1	12
z	4	0	4	0	4	0	1	33

All z-row coefficients $\geq 0 \rightarrow$ **no further improvement possible** $\rightarrow$ **OPTIMAL**.

Optimal Solution

Variable	Value
x1	0
x2	7
x3	0
x4	12
z*	33

Verification

$$\begin{aligned}
 z &= x1 + 3x2 - 4x3 + x4 \\
 &= 0 + 3(7) - 4(0) + 12 \\
 &= 21 + 12 \\
 &= 33 \checkmark
 \end{aligned}$$

Constraint 1: $x_1 + x_2 = 0 + 7 = 7 \leq 7 \quad \checkmark$ (tight)
 Constraint 2: $x_3 = 0 \leq 9 \quad \checkmark$
 Constraint 3: $x_1 - x_2 + x_4 = 0 - 7 + 12 = 5 \leq 5 \quad \checkmark$ (tight)

Intuition Behind the Solution

Variable	Why this value?
$x_2 = 7$	Coefficient +3 in objective (highest) → push to max. Constraint 1 limits it to 7.
$x_4 = 12$	Coefficient +1 in objective → push to max. With $x_2 = 7$, constraint 3 allows x_4 up to $0 - 7 + x_4 \leq 5$, so $x_4 \leq 12$.
$x_1 = 0$	Positive objective coefficient (+1), but entering x_1 would force x_2 or x_4 to decrease — net effect negative in the final tableau (reduced cost = -4).
$x_3 = 0$	Coefficient -4 in objective → increasing x_3 always hurts z ; never enters the basis.

Summary: Simplex Pivots

Iteration	Entering	Leaving	Pivot row	z value
Initial	—	—	—	0
1	x_2	s_1	Row 1	21
2	x_4	s_3	Row 3	33
Optimal	—	—	—	33

Exercise 2: EPON — IPACT Limited DBA

System Parameters

Parameter	Value
N (ONUs)	8
Distance OLT-ONU (d)	10 km
$W_{\max}$	10,000 bytes
Guard time (T_g)	2 μ s
Buffer size	1 MB
Max Ethernet frame	1,518 bytes
REPORT / GATE size	64 bytes
Upstream line rate (R)	2 Gbps

Parameter	Value
Traffic	Uniform load, memoryless (Poisson)

Pre-computed constants:

```

v = 300,000 km/s / 1.5 = 200,000 km/s    (speed in optical fiber)
RTT = 2 × 10 / 200,000 = 100 μs
T_wmax = 10,000 × 8 / 2×109 = 40 μs
T_ctrl = 64 × 8 / 2×109 = 0.256 μs    (REPORT or GATE transmission time)

```

(a) Minimal Cycle Length

The minimum possible cycle is RTT-limited: the OLT cannot send the next GATE until it has received the REPORT from the current cycle, which requires one full round trip.

```

T_cycle_min = RTT + T_report + T_gate
             = 100 + 0.256 + 0.256
             ≈ 100.5 μs

```

(b) Average Cycle Length at 95 % Load

Using the load-balance equation for IPACT Limited (Gated-style derivation):

- Cycle structure: $T = N \times (X + T_{\text{report}} + T_g)$ where X = data time per ONU
- Load definition: $L = N \times X / T \rightarrow N \times X = L \times T$
- Substituting: $T = L \times T + N \times (T_{\text{report}} + T_g) \rightarrow T(1-L) = N \times (T_{\text{report}} + T_g)$

```

T_cycle = N × (T_report + T_g) / (1 - L)
         = 8 × (0.256 + 2) / (1 - 0.95)
         = 8 × 2.256 / 0.05
         = 18.048 / 0.05
         ≈ 361 μs

```

Check: $361 \mu\text{s} > T_{\text{cycle_min}} = 100.5 \mu\text{s}$ ✓

(c) Average PON Packet Delay at 95 % Load

The average PON packet delay = queuing/waiting time at ONU + propagation from ONU to OLT.

Average waiting time in an IPACT polling system (packet arrives uniformly within the cycle):

$$W_{avg} = 1.5 \times T_{cycle} = 1.5 \times 361 = 541.5 \text{ }\mu\text{s}$$

One-way propagation delay:

$$T_{prop} = d / v = 10 / 200,000 = 50 \text{ }\mu\text{s}$$

Total average PON packet delay:

$$D_{avg} = W_{avg} + T_{prop} + T_{Wmax}/2 = 541.5 + 50 + 20 \approx 612 \text{ }\mu\text{s}$$

(d) Highest Achievable Load for $W_{max} = 5,000$ bytes

When W_{max} is capped at 5,000 bytes, every ONU can transmit at most W_{max} bytes per cycle. The system reaches saturation when all ONUs always fill their window:

$$T_{Wmax_new} = 5,000 \times 8 / 2 \times 10^9 = 20 \text{ }\mu\text{s}$$
$$L_{max} = T_{Wmax} / (T_{Wmax} + T_{report} + T_g)$$
$$= 20 / (20 + 0.256 + 2)$$
$$= 20 / 22.256$$
$$\approx 89.9 \%$$

Reducing W_{max} from 20,000 → 5,000 bytes lowers the maximum achievable load from ~97 % to ~89.9 %. A smaller window means the guard-time and control overhead represent a larger fraction of each slot, reducing efficiency.

Exercise 3: Wi-Fi 7 — Maximum Physical Bitrate

System Parameters

Parameter	Value
Modulation	4096-QAM → 12 bits/subcarrier
Coding rate	5/6
OFDM symbol time	12.8 μs
Guard Interval options	0.8 μs , 1.6 μs , 3.2 μs
OFDMA	4 × 980-tone RU (320 MHz band)

Parameter	Value
Spatial streams	8

(a) Maximum Physical Bitrate

For maximum bitrate, choose the **shortest Guard Interval = 0.8 μ s**.

Total OFDM symbol duration:

$$T_{\text{symbol}} = 12.8 + 0.8 = 13.6 \mu\text{s}$$

Total data subcarriers:

$$N_{\text{data}} = 4 \text{ RUs} \times 980 \text{ tones} = 3,920 \text{ subcarriers}$$

Bits per OFDM symbol (across all streams):

$$\begin{aligned} \text{bits/symbol} &= N_{\text{data}} \times \text{bits/subcarrier} \times \text{coding_rate} \times N_{\text{SS}} \\ &= 3,920 \times 12 \times (5/6) \times 8 \\ &= 3,920 \times 10 \times 8 \\ &= 313,600 \text{ bits/symbol} \end{aligned}$$

Maximum physical bitrate:

$$\begin{aligned} R_{\text{max}} &= \text{bits/symbol} / T_{\text{symbol}} \\ &= 313,600 / (13.6 \times 10^{-6}) \\ &\approx 23.06 \text{ Gbps} \end{aligned}$$

Guard Interval	T_{symbol}	Bitrate
0.8 μ s	13.6 μ s	23.06 Gbps ← maximum
1.6 μ s	14.4 μ s	21.78 Gbps
3.2 μ s	16.0 μ s	19.60 Gbps

Exercise 4: Wi-Fi Network Design

(a) Application Traffic Characteristics

Video stream:

- Application data per frame: 7,300 bytes + 12 bytes RTP header = 7,312 bytes
- UDP payload per IP packet: 1,500 – 20 (IP) – 8 (UDP) = 1,472 bytes
- IP packets per video frame: $\lceil 7,312 / 1,472 \rceil = 5$ IP packets
 - 4 × full: $20 + 8 + 1,472 = 1,500$ bytes
 - 1 × last: $20 + 8 + (7,312 - 4 \times 1,472) = 20 + 8 + 1,424 = 1,452$ bytes

Monitoring:

- IP packet: 72 (data) + 8 (UDP) + 20 (IP) = **100 bytes** → fits in one IP packet

Application	IP Packet Size	DL (AP → STA) / UL (STA → AP)	Traffic behavior
Video	1,500 bytes (max), 5 packets/frame	DL (AP → STA)	CBR, periodic at 50 fps
Monitoring	100 bytes	UL (STA → AP)	Periodic, 1 msg / 100 ms per STA

(b) T_{Data} Expression

For one PPDU carrying N aggregated packets of size P bytes at PHY rate R, using HT Greenfield format.

HT Greenfield preamble structure:

HT-GF-STF	HT-LTF 1	HT-SIG	HT-LTF 2 ... N_DLTF	DATA
8 μs	8 μs	8 μs	(N_DLTF-1) × 4 μs	

The number of data LTFs N_{DLTF} = smallest value in {1, 2, 4} ≥ N_{SS}:

N _{SS}	N _{DLTF}	Extra LTFs after SIG	Extra time
1	1	0	0 μs
2	2	1	4 μs
3 or 4	4	3	12 μs

This exercise specifies "No spatial multiplexing" → N_{SS} = 1 → N_{DLTF} = 1 → no extra HT-LTFs after HT-SIG.

Total preamble = 8 + 8 + 8 + 0 = **24 μs**

$$T_{\text{Data}} = 24 \mu\text{s} + (N \times P \times 8) / R$$

where P includes the full MPDU payload (IP + LLC + MAC header + FCS).

(c) Network Design

Step 1 — Channel Band

Choose 5 GHz.

Justification: 5 GHz has less interference than 2.4 GHz (fewer overlapping networks, less congestion from Bluetooth/microwave). Minimum MCS 3 (16-QAM, 1/2 coding) is guaranteed for all STAs.

Step 2 — Bandwidth

Choose 40 MHz.

From the MCS table, 5 GHz, MCS 3, Short GI, 40 MHz: $R = 60 \text{ Mbps}$.

This gives sufficient capacity with room for multiple streams and measurement STAs.

Step 3 — Timing verification (using T_{Data} formula)

Video DL — A-MPDU with 5 MPDUs per video frame:

$$\begin{aligned} P_{\text{MPDU}} &= \text{LLC}(8) + \text{IP}(1500) + \text{MAC_header}(24) + \text{FCS}(4) = 1,536 \text{ bytes} \\ \text{Total} &= 5 \times 1,536 = 7,680 \text{ bytes} \end{aligned}$$

$$\begin{aligned} T_{\text{Data}} &= 24 \mu\text{s} + 7,680 \times 8 / (60 \times 10^6) \\ &= 24 + 1,024 \\ &= 1,048 \mu\text{s} \approx 1.05 \text{ ms} \ll 200 \text{ ms delay budget} \checkmark \end{aligned}$$

Monitoring UL — one IP packet per poll, PCF polling:

$$P_{\text{MPDU}} = \text{LLC}(8) + \text{IP}(100) + \text{MAC_header}(24) + \text{FCS}(4) = 136 \text{ bytes}$$

$$T_{\text{Data}} = 24 + 136 \times 8 / (60 \times 10^6) = 24 + 18.1 = 42.1 \mu\text{s}$$

$$\begin{aligned} \text{Per STA: } T_{\text{POLL}} + \text{SIFS} + T_{\text{Data}} + \text{SIFS} &= 40 + 16 + 42.1 + 16 = 114.1 \mu\text{s} \\ 200 \text{ STAs: } 200 \times 114.1 &= 22,820 \mu\text{s} = 22.8 \text{ ms} < 100 \text{ ms delay budget} \checkmark \end{aligned}$$

Design Summary

Config	Choice	Justification
Channel	5 GHz	Less interference; MCS 3 guaranteed; suitable for high-rate short-range links
Bandwidth	40 MHz	60 Mbps at MCS 3; handles video + 200 monitoring STAs within delay budget
MAC mechanism	DCF (CSMA/CA + RTS/CTS) for video DL; PCF for monitoring UL	PCF polling ensures bounded UL delay for 200 STAs; RTS/CTS avoids hidden terminal for large video

Config	Choice	Justification
		frames
Aggregation	A-MPDU, 5 MPDUs per video PPDU	Amortises 24 μ s preamble over 5 packets; No ACK policy eliminates BlockACK overhead

(d) Saturation Check

Medium time consumed per second:

Video DL (with DIFS + backoff):

DIFS = $16 + 2 \times 9 = 34 \mu\text{s}$; avg backoff = $(15/2) \times 9 = 67.5 \mu\text{s}$

Per PPDU: $34 + 67.5 + 1,048 = 1,149.5 \mu\text{s}$

50 frames/s: $50 \times 1,149.5 = 57,475 \mu\text{s} = 57.5 \text{ ms/s}$

Monitoring UL (PCF, no contention overhead):

200 STAs polled every 100 ms $\rightarrow$ 10 rounds/s

Per round: $200 \times 114.1 \mu\text{s} = 22,820 \mu\text{s} = 22.8 \text{ ms}$

Per second: $10 \times 22.8 = 228 \text{ ms/s}$

Total used: $57.5 + 228 = 285.5 \text{ ms/s}$

Remaining capacity: $1,000 - 285.5 = 714.5 \text{ ms/s}$ ($\approx 71.5 \%$ free)

The network is NOT saturated.

Additional capacity available:

- Each extra video stream uses $\approx 57.5 \text{ ms/s}$ $\rightarrow$ room for **≈ 12 more streams**
- Each extra monitoring STA uses $\approx 10 \times 114.1 \mu\text{s} = 1.14 \text{ ms/s}$ $\rightarrow$ room for **hundreds more STAs**

Exercise 5: Cellular Networks — GSM & Spectrum Regulation

(a) Simultaneous Full-Rate Calls in One GSM Cell

Total physical channels:

$16 \text{ frequencies} \times 8 \text{ time slots/frequency} = 128 \text{ physical channels}$

Channels reserved for broadcast and common control:

5 time slots reserved for BCCH, FCCH, SCH, CCCH (PAGCH)

Channels needed for SDCCH (call setup signaling — 50 simultaneous setups required):

Each SDCCH/8 sub-channel uses 1 physical time slot and supports 8 parallel setups.

Number of SDCCH/8 slots = $\lceil 50/8 \rceil = 7 \text{ slots}$ ($7 \times 8 = 56 \geq 50$)

TCH channels available for active voice calls:

TCH = $128 - 5$ (broadcast/control) - 7 (SDCCH)
= 116 simultaneous full-rate calls

(b) Cellular vs. Wi-Fi — Spectrum Regulation & Radio Resource Management

	Cellular (4G/5G)	Wi-Fi
Spectrum regulation	Licensed, exclusive spectrum — operators purchase frequency bands at government auctions. Each operator owns their band; no one else can transmit in it.	Unlicensed, shared spectrum (ISM/UNII bands). Anyone can transmit without a licence, subject to power limits and etiquette rules (e.g., ETSI EN 300 328).
Radio resource management	Centralized: the base station (eNodeB/gNB) schedules every uplink and downlink resource block per slot. Guarantees QoS per user; inter-cell interference is managed by the network via frequency planning, power control, and coordination (X2 interface).	Distributed: CSMA/CA contention-based — each device decides independently when to transmit. No QoS guarantee (EDCA gives statistical priority only). Interference from neighboring APs/STAs is uncontrolled; adjacent deployments cannot coordinate.

Consequences:

- Cellular operators can guarantee service quality and plan coverage/capacity, but spectrum licensing is expensive → cost passed on to users via subscriptions.
- Wi-Fi is free to deploy but QoS degrades in dense environments where many devices/APs compete for the same unlicensed channel.

(c) "Cellular and Wi-Fi standards are converging"

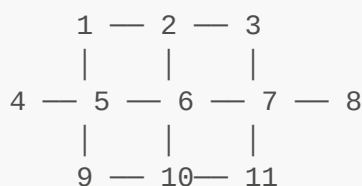
Several dimensions of convergence:

1. **Physical layer convergence:** Both now use OFDM/OFDMA modulation, MIMO, and beamforming. Wi-Fi 6 (802.11ax) borrowed OFDMA directly from LTE/5G; 5G NR adopted flexible numerology inspired by Wi-Fi's OFDM design.
2. **Feature convergence:** Cellular brought carrier aggregation and QoS bearers; Wi-Fi responded with EDCA (QoS), Block ACK, MU-MIMO (802.11ac/ax). 5G NR-U (New Radio Unlicensed) runs 5G protocols in Wi-Fi's unlicensed 5 GHz / 6 GHz bands.
3. **Spectrum convergence:** 5G is entering unlicensed bands (Wi-Fi territory), while private LTE/5G networks use spectrum that was once reserved for cellular operators only. The 6 GHz band is now shared between Wi-Fi 6E and private 5G deployments.
4. **Architecture convergence:** 3GPP defines tight WLAN–cellular interworking (EPC-integrated Wi-Fi offload, Passpoint/Hotspot 2.0, VoWiFi). Operators seamlessly hand traffic between cellular and Wi-Fi without the user noticing.
5. **Use case convergence:** Wi-Fi is used for voice calls (VoWiFi), video conferencing, and IoT — use cases once considered exclusively cellular. Cellular (NB-IoT, LTE-M) now targets low-power sensor deployments that were once only served by Zigbee/Wi-Fi.

In short: the boundary between "cellular" and "Wi-Fi" is blurring at the PHY, MAC, spectrum, and architecture levels.

Exercise 6: AODV in a Wireless Ad Hoc Network

Network Topology



Node 6 is the server. All 10 other nodes want to connect to Node 6.

Hop distances from Node 6:

Source node	Shortest path to 6	Hops (h)
1	6 → 5 → 1	2
2	6 → 2	1
3	6 → 7 → 3	2
4	6 → 5 → 4	2
5	6 → 5	1

Source node	Shortest path to 6	Hops (h)
7	$6 \rightarrow 7$	1
8	$6 \rightarrow 7 \rightarrow 8$	2
9	$6 \rightarrow 5 \rightarrow 9$	2
10	$6 \rightarrow 10$	1
11	$6 \rightarrow 7 \rightarrow 11$	2

(a) Worst-Case Total Routing Control Overhead

In AODV, for each route discovery:

- **RREQs:** $(N - 1) = 10$ broadcasts (every node except destination rebroadcasts once)
- **RREPs:** h unicast forwards back to the source (one per hop on the return path)

With **10 sources** connecting simultaneously (worst case = all at once, no caching):

Total RREQs = 10 sources $\times$ 10 RREQs each = 100 RREQ transmissions

Total RREPs = $\sum h_i = 2+1+2+2+1+1+2+2+1+2 = 16$ RREP transmissions

Total control overhead = 100 + 16 = 116 control packets

(b) Why Control Overhead Can Be Lower in Reality

1. **Route caching / intermediate RREP:** If a node already has a valid, unexpired routing table entry for Node 6 (learned from a previous RREQ flood), it can reply with an RREP immediately without forwarding the RREQ further. This cuts the RREQ flood short.
2. **Simultaneous setup reuse:** When 10 RREQs flood the network simultaneously, intermediate nodes that forward RREQ_A (from node 1) also record a reverse route back to node 1. Node 6 can then piggyback replies, and intermediate nodes gain routing knowledge for free.
3. **AODV route entry lifetime:** After the first connection is established, routing table entries remain valid for a TTL period. Subsequent connections from the same source reuse the cached route without re-flooding.
4. **Expanding ring search (TTL limiting):** If the destination is nearby, the RREQ is stopped before reaching all N nodes, dramatically reducing RREQs for short paths.

(c) Method Wi-Fi Uses to Avoid RREQ Collision

Wi-Fi uses **random jitter** before rebroadcasting: when a node receives a RREQ and decides to forward it, it waits a **random backoff delay** before actually transmitting. This desynchronises the rebroadcasts from

multiple neighbours, reducing the probability that two nodes transmit simultaneously and collide.

This is the standard IEEE 802.11 MAC collision-avoidance mechanism applied to broadcast RREQ propagation — a mitigation for the **broadcast storm problem**.

(d) Effect of Using Average Connection Speed as Route Metric

Standard AODV selects the route whose RREQ **arrives first** (minimum delay metric). Switching to **average connection speed** changes the procedure as follows:

1. **RREQ carries accumulated metric:** Each intermediate node appends the speed of the incoming link to the RREQ. The metric propagated is the minimum (bottleneck) link speed along the path so far.
2. **Destination cannot reply immediately:** Node 6 must wait to receive RREQs from multiple paths before it knows which path has the best metric. It sets a timer and waits for additional RREQs before sending the RREP.
3. **Multiple RREPs may be sent:** As better-metric RREQs arrive, Node 6 may send a new RREP superseding an earlier one. Intermediate nodes update their routing tables if the new RREP carries a better metric.
4. **Higher setup latency:** The connection setup is delayed because the destination waits for the best-metric RREQ rather than responding to the first one.
5. **Better path quality:** The selected route maximises bottleneck link speed, improving throughput at the cost of longer discovery time and potentially more control traffic.